

Contributions: In this paper, we make the following contributions to advance the state of the art. We describe the architecture and implementation of a *power-aware application placement framework*, called *pMapper*, which can incorporate various scenarios involving power and performance management using virtualization mechanisms. *pMapper* provides the solution to the most practical possibility, i.e. power minimization under performance constraints. We have implemented the framework, some algorithms, and the interfaces with an existing commercial IBM performance-oriented workload manager. We have benchmarked applications on virtualized server platforms to create utilization-based *power models* of application and server combinations and quantify the virtualization related costs. The characterization study provides us with insights into the structure of the power-aware placement problem that can be used to design tractable application placement solutions. We used the power models, migration cost models, and power-aware placement algorithms to design a dynamic placement controller that, under various assumptions, performs better than a static or load balancing placement controller with increasing heterogeneity of the server platforms, i.e. their power models. All input models and assumptions in *pMapper* have been validated on a testbed using a set of benchmark applications. The various algorithms implemented in *pMapper* have been compared through simulation on real utilization trace data obtained from a large production environment.

The rest of the paper is organized as follows. Section 2 describes the various flavors of the problem and the tool architecture. Section 3 discusses the assumptions made in the formulation and validates the assumptions through testbed experiments. Section 4 provides the details of the algorithms. Section 5 describes our implementation and a trace-driven evaluation study of our algorithms. Section 6 presents a comparative discussion with the related work.

2 pMapper: Power and Migration Cost-Aware Application Placement Framework

In this section, we present the *pMapper* application placement framework for power management and the various optimization formulations for power-aware application placement. We first present the *pMapper* architecture framework that leverages power management techniques enabled by virtualization.

2.1 Architecture

We have designed the *pMapper* framework to utilize all the power management capabilities available in virtualized platforms. As per the terminology used in [15], power management actions can be categorized as (i) soft actions like CPU idling in the hypervisor, (ii) hard actions like DVFS or throttling and (iii) consolidation actions. Commercial hypervisors drop all the power management actions that are taken by the OS. Further, for multi-tiered applications, a single VM instance may not be able to determine the application end-to-end QoS thus necessitating the need for a power management channel from the the management